

**Q: Why did you choose Logistic Regression?**

A: Because it is interpretable, stable, and widely used in financial systems. It provides probability outputs and works well with linearly separable data. In regulated domains like banking, interpretability is crucial.

**Q: Why not only accuracy?**

A: Accuracy can be misleading in imbalanced datasets. Our dataset had more rejections than approvals. So we used precision, recall, and F1-score for better evaluation.

**Q: Why is recall important in your project?**

A: Recall ensures we correctly identify deserving applicants. A low recall means rejecting genuine customers, which is a major loss for banks.

**Q: What is precision in your context?**

A: Precision ensures that when we approve a loan, it is actually safe. Low precision means giving loans to risky applicants.

**Q: Why did you use correlation heatmap?**

A: To understand relationships between features and target variable. It helped identify important features like Credit Score and DTI Ratio.

**Q: Why feature engineering?**

A: Because raw features had weak correlations. We created new features like squared terms and log transformation to capture non-linear patterns.

**Q: Why log transformation?**

A: Income data was skewed. Log transformation reduces skewness and improves model performance.

**Q: Why scaling?**

A: Scaling ensures all features are on the same range, which is important for models like KNN and Logistic Regression.

**Q: Why class\_weight='balanced'?**

A: Because dataset was imbalanced. This adjusts weights so the model does not favor majority class.

**Q: Why KNN performed worse?**

A: KNN is sensitive to noise and high dimensionality. It also depends heavily on distance, which may not capture complex patterns.

**Q: What is confusion matrix?**

A: It shows True Positives, True Negatives, False Positives, and False Negatives, giving a complete view of model performance.

**Q: What are false positives in your project?**

A: Approving a loan for a risky applicant.

**Q: What are false negatives?**

A: Rejecting a genuine applicant.

**Q: What improvements can you make?**

A: Use ensemble models like Random Forest or XGBoost, apply SMOTE, use real datasets, and add explainable AI.

**Q: What is overfitting?**

A: When model performs well on training data but poorly on unseen data.

**Q: How did you avoid data leakage?**

A: We applied scaling only on training data and then used it on test data.

**Q: What is deployment?**

A: Making the model usable in real-world through apps like Streamlit.

**Q: Why Streamlit?**

A: It is simple, fast, and allows quick deployment without frontend complexity.

**Q: Real-world application?**

A: Banks can use it for quick loan decisions, reducing manual work and bias.

**Q: What is your biggest learning?**

A: Understanding the importance of data preprocessing and feature engineering in improving model performance.